GATE CS — Computer Networks

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

Topic	Past questions	Total marks	New practice Qs
IP: addressing, CIDR, fragmentation, NAT	20	33	2
Data link: MAC, error detection, sliding window	17	32	2
Application layer, layering, DNS/HTTP, misc	22	28	2
TCP: flow & congestion control	14	22	2
Routing (DV/LS)	5	6	2
Total	**78**	**121**	**10**

How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

IP: addressing, CIDR, fragmentation, NAT

Subnetting/CIDR, forwarding tables, fragmentation, NAT. · 20 questions, 33 marks over 15 papers (≈2.2 marks/paper)

A. Past-paper questions (20)

2026 CS1 · Q.56 · 2 marks

An ISP having an address block 202.16.0.0/15 assigns a block of 6000 IP addresses to a client, using the classless internet domain routing (CIDR) super-netting approach. Which of the following address blocks can be assigned by the ISP?
(A) 202.16.0.0/19
(B) 202.17.64.0/19
(C) 202.16.32.0/19
(D) 202.17.24.0/19

2026 CS2 · Q.33 · 1 mark

If an IP network uses a subnet mask of 255.255.240.0, the maximum number of IP addresses that can be assigned to network interfaces is _____. (answer in integer)

2025 CS1 · Q.40 · 2 marks

A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?
(A) E3
(B) E1
(C) E2
(D) E5

2025 CS1 · Q.57 · 2 marks

Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header. The number of fragments that will be delivered to the destination is _____ . (Answer in integer)

2025 CS2 · Q.18 · 1 mark

A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X. Which ONE of the following is NOT a possible candidate for X?
(A) Internet Control Message Protocol (ICMP)
(B) Internet Group Management Protocol (IGMP)
(C) Open Shortest Path First (OSPF)
(D) Routing Information Protocol (RIP)

2025 CS2 · Q.23 · 1 mark
Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?
(A) 5 fragments, 1488 bytes
(B) 6 fragments, 88 bytes
(C) 6 fragments, 108 bytes
(D) 6 fragments, 116 bytes

2024 CS1 · Q.31 · 1 mark
Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?
(A) Source IP
(B) Destination IP
(C) Header Checksum
(D) Total Length

2024 CS1 · Q.58 · 2 marks
Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation “/N” in a prefix indicates a subnet mask with the most significant N bits set to 1. Prefix Next hop router 10.1.1.0/24 R1 10.1.1.128/25 R2 10.1.1.64/26 R3 10.1.1.192/26 R4 This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205 . The number of packets forwarded via the next hop router R2 is _____

2024 CS1 · Q.65 · 2 marks
Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____

2024 CS2 · Q.28 · 1 mark
Which of the following statements about IPv4 fragmentation is/are TRUE? The fragmentation of an IP datagram is performed only at the source of the
(A) datagram The fragmentation of an IP datagram is performed at any IP router which finds
(B) that the size of the datagram to be transmitted exceeds the MTU
(C) The reassembly of fragments is performed only at the destination of the datagram The reassembly of fragments is performed at all intermediate routers along the
(D) path from the source to the destination

2024 CS2 · Q.32 · 1 mark
Which of the following fields of an IP header is/are always modified by any router before it forwards the IP packet?
(A) Source IP Address
(B) Protocol
(C) Time to Live (TTL)
(D) Header Checksum

2024 CS2 · Q.38 · 2 marks
IP addresses 10.12.2.0 to 10.12.3.255?
(A) 10.12.2.0/23
(B) 10.12.2.0/24
(C) 10.12.0.0/22
(D) 10.12.2.0/22

2023 · Q.65 · 2 marks
The forwarding table of a router is shown below. Subnet Number Subnet Mask Interface ID 200.150.0.0 255.255.0.0 1 200.150.64.0 255.255.224.0 2 200.150.68.0 255.255.255.0 3 200.150.68.64 255.255.255.224 4 Default 0 A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID . CS

2022 · Q.55 · 2 marks
Consider routing table of an organization’s router shown below: Subnet Number Subnet Mask Next Hop 12.20.164.0 255.255.252.0 R1 12.20.170.0 255.255.254.0 R2 12.20.168.0 255.255.254.0 Interface 0 12.20.166.0 255.255.254.0 Interface 1 default R3 Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?
(A) 12.20.164.0/20
(B) 12.20.164.0/22
(C) 12.20.164.0/21
(D) 12.20.168.0/22

2021 · Q.45 · 2 marks [OCR]
Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes. A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as 0x1234. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., Don’t Fragment (DF) flag in the IP header is not set by P. Which of the following statements is/are correct?
(A) | Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.
(B) | If the second fragment is lost, R will resend the fragment with the IP identification value 0x1234.
(C) | If the second fragment is lost, P is required to resend the whole TCP segment.
(D) | TCP destination port can be determined by analysing only the second fragment. CS, Set-1 - Copyright © GATE 2021

2020 · Q.38 · 2 marks [OCR]
An organization requires a range of IP addresses to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one to the organization? I. 202.61.84.0/21 Il. — 202.61.104.0/21 Il. 202.61.64.0/21 IV. 202.61.144.0/21
(A) land II only (8) Il and III only () Ill and IV only (0) land IV only

2019 · Q.28 · 2 marks [OCR]
Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?
(A) M,N, and P all belong to the same subnet
(B) Only M and N belong to the same subnet
(C) Only N and P belong to the same subnet
(D) M,N, and P belong to three different subnets

2018 · Q.54 · 2 marks
Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0. The fragmentation offset value stored in the third fragment is _____. CS 19/20 GATE 2018

2017 CS2 · Q.20 · 1 mark [OCR]
The maximum number of [Pyv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is =====PAGE 9=====

2016 · Q.53 · 2 marks
An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is .

B. New practice questions with solutions

1. (1 mark · NAT) A network uses the subnet mask 255.255.255.192. The number of usable host addresses per subnet is _____.

62. The mask is /26, leaving 6 host bits, so $2^6 - 2 = 62$.

2. (2 marks · MCQ) A router has these forwarding entries: 10.0.0.0/8 → I1, 10.20.0.0/16 → I2, 10.20.32.0/19 → I3, 10.20.33.0/24 → I4, default → I5. A packet for 10.20.34.7 leaves through

A. I1 (B) I2 (C) I3 (D) I4

(C). The /19 entry covers 10.20.32.0 to 10.20.63.255, which includes 10.20.34.7. The /24 entry

> (10.20.33.x) does not match. Longest prefix match picks the /19.

Data link: MAC, error detection, sliding window

Sliding window / stop-and-wait, CRC/Hamming, CSMA/Ethernet. · 17 questions, 32 marks over 15 papers (≈2.1 marks/paper)

A. Past-paper questions (17)

2026 CS1 · Q.45 · 2 marks

Consider the implementation of sliding window protocol over a lossless link, with a window size of W frames, where each frame is of size 1000 bits (including header). The bandwidth of the link is 100 kbps (1k = 103) and the one-way propagation delay is 100 milliseconds. Assume that processing times at the sender and receiver are zero and the transmission time of acknowledgements is also zero. Which one of the following options gives the minimum size of W (in number of frames) required to achieve 100% link utilization?
(A) 10
(B) 21
(C) 20
(D) 11

2026 CS2 · Q.43 · 2 marks

Consider the transmission of data bits 110001011 over a link that uses Cyclic Redundancy Check (CRC) code for error detection. If the generator bit pattern is given to be 1001, which one of the following options shows the remainder bit pattern appended to the data bits before transmission?
(A) 011
(B) 101
(C) 000
(D) 100

2026 CS2 · Q.65 · 2 marks

It is necessary to design a link-layer protocol between two hosts that are directly connected over a lossless link of length 3000 kilometers. Assume that the link bandwidth is 108 bits per second and that the propagation delay in the link is 5 nanoseconds per meter. Every transmitted data byte is assigned a unique sequence number. Let N be the minimum number of bits needed for the sequence number field in the protocol header such that i. the sequence numbers do not wrap around before 60 seconds, and ii. the maximum utilization of the link is achieved. The value of N is _____. (answer in integer)

2025 CS2 · Q.36 · 2 marks

Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?
(A) 88.23
(B) 93.75
(C) 85.44
(D) 66.67

2024 CS2 · Q.23 · 1 mark

Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection. Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

(A) The destination IP address is the IP address of R

(B) The destination IP address is the IP address of Y

(C) The destination MAC address is the MAC address of S

(D) The destination MAC address is the MAC address of Y

2024 CS2 · Q.55 · 2 marks

Consider an Ethernet segment with a transmission speed of 108 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×108 meters/sec, then the minimum frame size (in bits) required for collision detection is _____

2023 · Q.17 · 1 mark

Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.

(A) Longer link length and lower transmission rate

(B) Longer link length and higher transmission rate

(C) Shorter link length and lower transmission rate

(D) Shorter link length and higher transmission rate CS

2022 · Q.59 · 2 marks

Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3x108 m/s. The time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is_____.

2021 · Q.29 · 2 marks [OCR]

Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is dgdzdgdscgdyd4d3dyce4d, cc, where the data bits and the check bits are given in the following tables: Data bits Check bits ds | dz | de | ds | ds | ds | de | dh es | ca | [er 1/1/0/;/a¢]O0};1]0}1 y{O}1]0 Which one of the following choices gives the correct values of x and y?

(A) | x is 0 and y is 0.

(B) | z is 0 and y is 1.

(C) | x is 1 and y is 0.

(D) | zis 1 and y is 1. CS, Set-1 - Copyright © GATE 2021

2021 · Q.49 · 2 marks [OCR]
Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link. Assume the following: e The time taken for processing the data frame by the receiver is negligible. e The time taken for processing the acknowledgement frame by the sender is neg- ligible. e The sender has infinite number of frames available for transmission. e The size of the data frame is 2,000 bits and the size of the acknowledgement frame is 10 bits. e The link data rate in each direction is 1 Mbps (= 10° bits per second). e One way propagation delay of the link is 100 milliseconds. The minimum value of the sender’s window size in terms of the number of frames, (rounded to the nearest integer) needed to achieve a link utilization of 50% is

2019 · Q.49 · 2 marks [OCR]
Consider that 15 machines need to be connected in a LAN using 8-port Ethemet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is

2018 · Q.55 · 2 marks
Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium. Assume that the system has two nodes P and Q, located at a distance d meters from each other. P starts transmitting a packet at time t=0 after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time t=0 and begins to carrier-sense the medium. The maximum distance d (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P’s ongoing transmission is _____. END OF THE QUESTION PAPER CS 20/20 Q.No. Type Section Key/Range Marks 1 MCQ GA B 1 2 MCQ GA A 1 3 MCQ GA D 1 4 MCQ GA C 1 5 MCQ GA B 1 6 MCQ GA B 2 7 MCQ GA C 2 8 MCQ GA B 2 9 MCQ GA A 2 10 MCQ GA C 2 1 MCQ CS D 1 2 MCQ CS A 1 3 MCQ CS B 1 4 MCQ CS D 1 5 MCQ CS D 1 6 MCQ CS D 1 7 MCQ CS B 1 8 MCQ CS B 1 9 MCQ CS A 1 10 MCQ CS B 1 11 MCQ CS A 1 12 MCQ CS D 1 13 MCQ CS C 1 14 MCQ CS C 1 Q.No. Type Section Key/Range Marks 15 NAT CS 0.021 to 0.024 1 16 NAT CS 0.27 to 0.30 1 17 NAT CS 3 to 3 1 18 NAT CS 3 to 3 1 19 NAT CS 42 to 42 1 20 NAT CS 4 to 4 1 21 NAT CS 4 to 4 1 22 NAT CS 2 to 2 1 23 NAT CS 59.0 to 60.0 1 24 NAT CS 2 to 2 1 25 NAT CS 34 to 35 1 26 MCQ CS D 2 27 MCQ CS D 2 28 MCQ CS A 2 29 MCQ CS A 2 30 MCQ CS A 2 31 MCQ CS C 2 32 MCQ CS B 2 33 MCQ CS C 2 34 MCQ CS B 2 35 MCQ CS B 2 36 MCQ CS D 2 37 MCQ CS D 2 38 MCQ CS A 2 Q.No. Type Section Key/Range Marks 39 MCQ CS A 2 40 MCQ CS C 2 41 MCQ CS C 2 42 MCQ CS B 2 43 NAT CS 109 to 109 2 44 NAT CS 0.60 to 0.62 2 45 NAT CS 10230 to 10230 2 46 NAT CS 80 to 80 2 47 NAT CS 4 to 4 2 48 NAT CS 16 to 16 2 49 NAT CS 3 to 3 2 50 NAT CS 219 to 219 2 51 NAT CS 32 to 32 2 52 NAT CS 2 to 2 2 53 NAT CS 85 to 85 2 54 NAT CS 144 to 144 2 55 NAT CS 50 to 50 2

2017 CS1 · Q.32 · 2 marks [OCR]
A computer network uses polynomials over GF(2) for error checking with 8 bits as information bits and usesx° +x +1 as the generator polynomial to generate the check bits. In this network. the message 01011011 is transmitted as
(A) 01011011010
(B) 01011011011
(C) 01011011101
(D) 01011011100

2017 CS1 · Q.45 · 2 marks [OCR]
The values of parameters for the Stop-and-Wait ARQ protocol are as given below: Bit rate of the transmission channel = 1 Mbps. Propagation delay from sender to receiver = 0.75 ms. Time to process a frame = 0.25 ms. Number of bytes in the information frame = 1980. Number of bytes in the acknowledge frame = 20. Number of overhead bytes in the information frame = 20. Assume that there are no transmission errors. Then. the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is (correct to 2 decimal places). =====PAGE 21=====

2017 CS2 · Q.34 · 2 marks [OCR]
Consider a binary code that consists of only four valid codewords as given below: 00000, 01011, 10101, 11110 Let the minimum Hamming distance of the code be p and the maximum number of erroneous bits that can be corrected by the code be g. Then the values of p and g are
(A) p=3 and g=1
(B) p=3 and g=2
(C) p=4 and g=1
(D) p=4 and g=2

2017 CS2 · Q.35 · 2 marks [OCR]
Consider two hosts X¥ and Y, connected by a single direct link of rate 10° bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2 x 108 m/sec. Host X sends a file of 50,000 bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and g milliseconds, respectively. Then the values of p and g are
(A) p=50 and g=100
(B) p=50 and g=400
(C) p=100 and g=50
(D) p=400 and g=50 =====PAGE 14=====

2016 · Q.55 · 2 marks
A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is bytes/second.

B. New practice questions with solutions

1. (1 mark · NAT) Stop-and-wait is used on a 1 Mbps link with 1000-bit frames and a 4 ms one-way propagation delay. Ignoring ACK transmission time, the link efficiency in percent, to

one decimal place, is _____.

> **11.1.** T_t = 1 ms and the round trip adds 8 ms. Efficiency = 1 / (1 + 8) = 11.1%.

2. (2 marks · MCQ) Data 1101 is sent with a CRC using generator polynomial $x^3 + x + 1$ (bits 1011). The transmitted codeword is

A. 1101001 (B) 1101011 (C) 1101100 (D) 1101110

> **(A).** Dividing 1101000 by 1011 over GF(2) leaves remainder 001, which is appended to the data.

Application layer, layering, DNS/HTTP, misc

Layering, DNS/HTTP, ARP/DHCP, delay and bandwidth, (a few security questions). · 22 questions, 28 marks over 15 papers (≈1.9 marks/paper)

A. Past-paper questions (22)

2026 CS1 · Q.19 · 1 mark

Which of the following statements is/are true with respect to the interaction of a web browser with a web server using HTTP 1.1?
(A) HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, if the objects are stored in the same server
(B) HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, even if they are stored in different servers
(C) HTTP 1.1 facilitates sending a request for downloading one object without waiting for a previously requested object to be downloaded completely
(D) HTTP 1.1 facilitates downloading multiple webpages on the same server to be downloaded over a single TCP connection

2026 CS2 · Q.21 · 1 mark

connected via a path consisting of three consecutive links of bandwidth 2 Mbps, 500 kbps, and 1 Mbps, respectively. All processing delays and propagation delays are negligible. Assume that there is no other background traffic over the path and no other additional overhead to transfer the file. Which one of the following is the total time (in seconds) to transfer the file? Note: 1M=106, 1k=103
(A) 731
(B) 64
(C) 8
(D) 16

2026 CS2 · Q.22 · 1 mark

Which one of the following protocols may need to broadcast some of its messages?
(A) SMTP
(B) FTP
(C) DHCP
(D) HTTP

2025 CS1 · Q.16 · 1 mark

Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown. OSI Layers Functionalities (a) Network layer (I) Packet routing (b) Transport layer (II) Framing and error handling (c) Datalink layer (III) Host to host communication
(A) (a)-(I), (b)-(II), (c)-(III)
(B) (a)-(I), (b)-(III), (c)-(II)
(C) (a)-(II), (b)-(I), (c)-(III)
(D) (a)-(III), (b)-(II), (c)-(I)

2025 CS2 · Q.16 · 1 mark

Consider the following statements: (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address. (ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S. Which ONE of the following is CORRECT?

(A) Both (i) and (ii) are TRUE
(B) (i) is TRUE and (ii) is FALSE
(C) (i) is FALSE and (ii) is TRUE
(D) Both (i) and (ii) are FALSE

2024 CS1 · Q.16 · 1 mark

A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order. (i) HTTP GET request for the index page (ii) DNS request to resolve the web server's name to its IP address (iii) HTTP GET request for an image object (iv) TCP SYN to open a connection to the web server Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ?

(A) (iv), (ii), (iii), (i)
(B) (ii), (iv), (iii), (i)
(C) (ii), (iv), (i), (iii)
(D) (iv), (ii), (i), (iii)

2024 CS1 · Q.36 · 2 marks

sends a file of size 106 bytes to R via this path by splitting the file into chunks of 103 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU. Each of the links P—Q and Q—R has a bandwidth of 106 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link P—Q and transmit on link Q —R. Assume P starts transmitting the chunks at time t = 0. Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

(A) 8.000
(B) 8.008
(C) 15.992
(D) 16.000

2023 · Q.52 · 2 marks
Suppose in a web browser, you click on the www.gate-2023.in URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers. Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting www.gate-2023.in is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server. Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?
(A) 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(B) 5 RTTs, in case of persistent HTTP with pipelining.
(C) 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(D) 6 RTTs, in case of persistent HTTP with pipelining. CS

2022 · Q.22 · 1 mark
Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below. What is the number of subnets inside the enterprise network?
(A) 3
(B) 12
(C) 6
(D) 8

2022 · Q.35 · 1 mark
Consider the resolution of the domain name www.gate.org.in by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is_____.

2021 · Q.8 · 1 mark [OCR]
Consider the following two statements. Sy: Destination MAC address of an ARP reply is a broadcast address. So: Destination MAC address of an ARP request is a broadcast address. Which one of the following choices is correct?
(A) | Both S; and Sy are true.
(B) | S; is true and Sy is false.
(C) | S; is false and S) is true.
(D) | Both S; and 52 are false. CS, Set-1 - Copyright © GATE 2021

2020 · Q.25 · 1 mark [OCR]
Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is Q26 - Q55 carry two marks each.

2019 · Q.16 · 1 mark [OCR]
Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?
(A) IMAP, POP3
(B) SMTP, POP3
(C) SMTP, MIME
(D) IMAP, SMTP cs 3/15 GATE 2019 Computer Science and Information Technology Set-2

2019 · Q.29 · 2 marks [OCR]
Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?
(A) X sends an ARP request packet to the local gateway’s IP address which then finds the MAC address of Y and sends to X
(B) X sends an ARP request packet to the local gateway’s MAC address which then finds the MAC address of Y and sends to X
(C) X sends an ARP request packet with broadcast MAC address in its local subnet
(D) X sends an ARP request packet with broadcast IP address in its local subnet

2019 · Q.54 · 2 marks [OCR]
In an RSA cryptosystem, the value of the public modulus parameter n is 3007. If it is also known that @(n) = 2880, where ¢() denotes Euler’s Totient Function, then the prime factor of n which is greater than 50 is .

2018 · Q.13 · 1 mark
Match the following: Field Length in bits P. UDP Header’s Port Number I. 48 Q. Ethernet MAC Address II. 8 R. IPv6 Next Header III. 32 S. TCP Header’s Sequence Number IV. 16
(A) P-III, Q-IV, R-II, S-I
(B) P-II, Q-I, R-IV, S-III
(C) P-IV, Q-I, R-II, S-III
(D) P-IV, Q-I, R-III, S-II CS 4/20 GATE 2018

2017 CS1 · Q.15 · 1 mark [OCR]
A sender S sends a message 7 to receiver R. which is digitally signed by S with its private key. In this scenario. one or more of the following security violations can take place. () Scan launch a birthday attack to replace TM with a fraudulent message. (II) A third party attacker can launch a birthday attack to replace 7 with a fraudulent message. (II) R can launch a birthday attack to replace TM with a fraudulent message. Which of the following are possible security violations?
(A) (and (ID only
(B) (only
(C) (I) only
(D) (II) and (IID) only

2017 CS1 · Q.44 · 2 marks [OCR]
In a RSA cryptosystem. a participant A uses two prime numbers p= 13 and g =17 to generate her public and private keys. If the public key of A is 35, then the private key of A is

2017 CS2 · Q.18 · 1 mark [OCR]
Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT? Ti A connected UDP socket can be used to communicate with multiple peers simultaneously. IL. A process can successfully call connect function again for an already connected UDP socket.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II =====PAGE 8=====

2016 · Q.24 · 1 mark
Which one of the following protocols is NOT used to resolve one form of address to another one?
(A) DNS
(B) ARP
(C) DHCP
(D) RARP

2016 · Q.25 · 1 mark
Which of the following is/are example(s) of stateful application layer protocols? (i) HTTP (ii) FTP (iii) TCP (iv) POP3
(A) (i) and (ii) only
(B) (ii) and (iii) only
(C) (ii) and (iv) only
(D) (iv) only

2016 · Q.52 · 2 marks
Consider that B wants to send a message m that is digitally signed to A. Let the pair of private and public keys for A and B be denoted by Kx− and Kx+ for x = A, B, respectively. Let Kx (m) represent the operation of encrypting m with a key Kx and H(m) represent the message digest. Which one of the following indicates the CORRECT way of sending the message m along with the digital signature to A?
(A) {m, KB+ (H(m))}
(B) {m, KB− (H(m))}
(C) {m, KA− (H(m))}
(D) {m, KA+ (m)}

B. New practice questions with solutions

1. (1 mark · MCQ) Which application protocol and default transport pairing is correct?

A. SMTP over UDP
B. HTTP/1.1 over UDP
C. DNS queries over UDP
D. FTP over UDP

(C). DNS queries normally use UDP, with TCP used for large responses and zone transfers. SMTP, HTTP/1.1 and FTP use TCP.

2. (2 marks · NAT) A message of 10 packets of 1000 bits each crosses three 1 Mbps links in series, with two store-and-forward routers and negligible propagation delay. The total time to deliver the whole message in milliseconds is _____.

> **12.** The first packet needs 3 ms to cross the path, and each remaining packet arrives 1 ms after the previous one. 3 + 9 = 12 ms, which equals (10 + 3 − 1) × 1 ms.

TCP: flow & congestion control

Congestion window, handshake, sequence numbers, throughput. · 14 questions, 22 marks over 15 papers (≈1.5 marks/paper)

A. Past-paper questions (14)

2026 CS1 · Q.18 · 1 mark

With respect to a TCP connection between a client and a server, which one of the following statements is true?
(A) The client and server use a two-way handshake mechanism before the start of data transmission
(B) The server cannot initiate closing of the connection before the client initiates closing of the connection
(C) The TCP connection is half-duplex
(D) The client and server can initiate closing of the connection at the same time

2026 CS1 · Q.44 · 2 marks

A TCP sender successfully establishes a connection with a TCP receiver and starts the transmission of segments. The TCP congestion control mechanism’s slow-start threshold is set to 10000 segments. Assume that the round-trip time is fixed at 1 millisecond. Assume that the sender always has data to send, the segments are numbered from 1, and no segment is lost. Let t denote the time (in milliseconds) at which the transmission of segment number 2000 starts. Which one of the following options is correct?
(A) $9 \leq t < 10$
(B) $10 \leq t < 11$
(C) $11 \leq t < 12$
(D) $12 \leq t < 13$

2026 CS2 · Q.58 · 2 marks

Consider a new TCP connection between a sender and a receiver. The receiver advertised window is constant at 48 KB, the maximum segment size (MSS) is 2 KB, and the slow start threshold for TCP congestion control is 16 KB. Assume that there are no timeouts or duplicate acknowledgements. The number of rounds of transmission required for the congestion control algorithm of the TCP connection to reach the congestion avoidance phase is _____. (answer in integer) Note: 1K=210

2025 CS1 · Q.22 · 1 mark

Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?
(A) P3: SYN = 1, ACK = 1
(B) P2: SYN = 1, ACK = 1
(C) P2: SYN = 0, ACK = 1
(D) P1: SYN = 1

2024 CS1 · Q.29 · 1 mark

TCP client P successfully establishes a connection to TCP server Q. Let NP denote the sequence number in the SYN sent from P to Q. Let NQ denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

(A) The sequence number NP is chosen randomly by P

(B) The sequence number NP is always 0 for a new connection

(C) The acknowledgement number NQ is equal to NP

(D) The acknowledgement number NQ is equal to NP + 1

2024 CS2 · Q.54 · 2 marks

Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____

2023 · Q.50 · 2 marks

Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

(A) 30 bits

(B) 32 bits

(C) 34 bits

(D) 36 bits CS

2022 · Q.60 · 2 marks

Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is_____.

2021 · Q.44 · 2 marks [OCR]

A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

(A) | If the client was waiting to receive a packet, it may wait indefinitely.

(B) | The TCP server application on S can listen on P after reboot.

(C) | If the client sends a packet after the server reboot, it will receive a RST segment.

(D) | If the client sends a packet after the server reboot, it will receive a FIN segment.

2020 · Q.55 · 2 marks [OCR]

Consider a TCP connection between a client and a server with the following specifications: the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time t = 0. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time t + 60 ms after all acknowledgements are processed is Copyright : GATE 2020, IIT Delhi GATE 2020 Graduate Aptitude Test in Engineering 2020 Home Information Brochure GATE International Pre Examination Answer Key - CS: Computer Sc. and Information Technology IIT Delhi Organising Institute Important Dates QNo. [Session | Que. Type |Sec. Name |Key Marks 1 6 cq GA D 1 2 6 cq GA A 1 3 6 cq GA A 1 4 6 McQ GA D 1 5 6 cq GA B 1 6 6 cq GA D 2 7 6 McQ GA c 2 8 6 McQ GA A 2 9 6 cq GA B 2 10 6 cq GA c 2 1 6 McQ cs A 1 2 6 McQ cs A 1 3 6 cq cs c 1 4 6 cq cs c 1 5 6 McQ cs B 1 6 6 McQ cs c 1 7 6 cq cs MTA 1 8 6 cq cs D 1 9 6 McQ cs D 1 10 6 cq cs c 1 "1 6 cq cs c 1 12 6 cq cs c 1 B 6 McQ cs A 1 14 6 cq cs A 1 15 6 cq cs D 1 16 6 McQ cs c 1 17 6 NAT cs 0.125 to 0.125 1 18 6 NAT cs 7to7 1 19 6 NAT cs 5to5 1 20 6 NAT cs 1034 to 1034 1 n 6 Nat G 13.3 to 13.3 OR 13.5 to 13.5 2 6 NAT cs 19 to 19 1 23 6 NAT cs 13 to 13 1 24 6 NAT cs 7to7 1 25 6 NAT cs 6to6 1 26 6 cq cs A 2 7 6 McQ cs c 2 28 6 cq cs B 2 29 6 cq cs B 2 30 6 cq cs B 2 31 6 McQ cs D 2 32 6 cq cs A 2 FAQs Contact Us 33 MCQ cs B 34 MCQ cs A 35 MCQ cs B 36 MCQ cs A 37 MCQ cs A 38 MCQ cs B 39 MCQ cs Cc 40 MCQ cs A 4 MCQ cs B 42 NAT cs 12 to 12 43 NAT cs 2.15 to 2.18 44 NAT cs 14 to 14 45 NAT cs 0.5 to 0.5 46 NAT cs 55 to 55 47 NAT cs 511 to 511 48 NAT cs 81 to 81 49 NAT cs 99 to 99 50 NAT cs 5.25 to 5.25 51 NAT cs 6to6 52 NAT cs 7to7 53 NAT cs 154.5 to 155.5 54 NAT cs 4to4 55 NAT cs 44 to 44 Copyright : GATE 2020, IIT Delhi

2018 · Q.14 · 1 mark

Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that cwnd stands for the TCP congestion window and MSS denotes the Maximum Segment Size. (i) The cwnd increases by 2 MSS on every successful acknowledgment. (ii) The cwnd approximately doubles on every successful acknowledgement. (iii) The cwnd increases by 1 MSS every round trip time. (iv) The cwnd approximately doubles every round trip time. Which one of the following is correct?

(A) Only (ii) and (iii) are true
(B) Only (i) and (iii) are true
(C) Only (iv) is true
(D) Only (i) and (iv) are true

2018 · Q.25 · 1 mark

Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps (= 109 bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

2017 CS1 · Q.14 · 1 mark [OCR]
Consider a TCP client and a TCP server running on two different machines. After completing data transfer. the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK. which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793). in which state does the client-side TCP connection wait for the FIN from the server-side TCP?
(A) LAST-ACK
(B) TIME-WAIT
(C) FIN-WAIT-1
(D) FIN-WAIT-2 =====PAGE 9=====

2016 · Q.54 · 2 marks
For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is seconds.

B. New practice questions with solutions

1. (1 mark · NAT) A TCP sender starts with cwnd = 1 MSS in round 1 and ssthresh = 16 MSS, with no losses. Slow start doubles cwnd each round until it reaches ssthresh, then congestion avoidance adds 1 MSS per round. The cwnd in round 8 is _____ MSS.

19. The values are 1, 2, 4, 8, 16 in rounds 1–5, then 17, 18 and 19 in rounds 6–8.

2. (2 marks · NAT) A TCP connection has a fixed 64 KB window (1 KB = 1024 bytes), a 40 ms RTT and a 100 Mbps link. Ignoring all other effects, the link utilisation in percent, to one decimal place, is _____.

13.1. Throughput is at most window/RTT = 524,288 bits / 0.04 s = 13.1 Mbps, which is 13.1% of 100 Mbps.

Routing (DV/LS)

Distance-vector, link-state, OSPF. · 5 questions, 6 marks over 15 papers (≈0.4 marks/paper)

A. Past-paper questions (5)

2025 CS2 · Q.17 · 1 mark
Consider the routing protocols given in List I and the names given in List II: List I List II (i) Distance vector routing (a) Bellman-Ford (ii) Link state routing (b) Dijkstra For matching of items in List I with those in List II, which ONE of the following options is CORRECT?
(A) (i) – (a) and (ii) – (b)
(B) (i) – (a) and (ii) – (a)
(C) (i) – (b) and (ii) – (a)
(D) (i) – (b) and (ii) – (b)

2023 · Q.25 · 1 mark
Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?
(A) OSPF implements Bellman-Ford algorithm to find shortest paths.
(B) OSPF uses Dijkstra’s shortest path algorithm to implement least-cost path routing.
(C) OSPF is used as an inter-domain routing protocol.
(D) OSPF implements hierarchical routing. CS

2022 · Q.57 · 2 marks
Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity. P Q R The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q−R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to count-to-infinity problem, is_____.

2020 · Q.15 · 1 mark [OCR]
Consider the following statements about the functionality of an IP based router. I. A router does not modify the IP packets during forwarding. II. It is not necessary for a router to implement any routing protocol. III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet. Which of the above statements is/are TRUE? (a) land II only (8) lTonly (©) IL and II only (0) Il only

2017 CS2 · Q.9 · 1 mark [OCR]
Consider the following statements about the routing protocols. Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network. I: RIP uses distance vector routing II: RIP packets are sent using UDP Ill: OSPF packets are sent using TCP IV: OSPF operation is based on link-state routing Which of the statements above are CORRECT?
(A) Land IV only
(B) I. IL and III only
(C) I. ILand IV only
(D) I, III and IV only =====PAGE 5=====

B. New practice questions with solutions

1. (1 mark · NAT) Router X has neighbours A (link cost 2) and B (link cost 5). A advertises a cost of 7 to destination D and B advertises a cost of 3. X’s distance-vector cost to D is _____.

> **8.** min(2 + 7, 5 + 3) = 8.

2. (2 marks · NAT) Link costs are R1–R2 = 2, R1–R3 = 5, R2–R3 = 1, R2–R4 = 7, R3–R4 = 3, R4–R5 = 2 and R3–R5 = 8. Using Dijkstra from R1, the sum of the shortest distances to R2, R3, R4 and R5 is _____.

> **19.** The distances are R2 = 2, R3 = 3, R4 = 6 and R5 = 8, and they sum to 19.